

ARM PrimeCell Driver

SSMC v1.0 (PL093)
Test Report

PrimeCell Driver Test Report

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Document Summary

Document Issue	A
PrimeCell driver identifier	SSMC
Code Version	1.0
PrimeCell identifier	PL093
Test date	19, March, 2002

Contents

1	ABOUT THIS DOCUMENT	5
1.1	Change history	5
2	RELEASED FILES	6
3	TEST ENVIRONMENT	6
3.1	Test Specification Summary	6
3.2	Hardware requirements	7
4	CODE COVERAGE	8
4.1	Overall	8
4.2	Untested Blocks	8
4.3	Untested Routine Exits	9
5	RELEASED TEST MODULE	10
5.1	Overview	10
5.2	Changes to test environment	10
5.3	Configuration of the test module	10
5.4	Outline	10
5.5	Results	11
5.6	Performance obtained	11

1 ABOUT THIS DOCUMENT

1.1 Change history

Issue	Date	Change
A	19, March, 2002	First release of test report

2 RELEASED FILES

PL093-ISPC-1000-A	API document
PL093-TREP-1000	Test Results document
apPL093/apPL093.h	Public header file
apPL093/PL093.h	Private header file
apPL093/PL093.c	Source code
apPL093/PL093tst.c	Released test code

3 TEST ENVIRONMENT

3.1 Test Specification Summary

	Specification (Intended)	Actual Test Environment If different from Specification
Code Test Platform	INTEGRATOR	
Core used for test	ARM7TDMI	ARM 940T
Module ROM size (RO code + RO data)	1224 Bytes	
Module RAM size (ZI data) per instance	8 Bytes	
PrimeCell tested	PL093	
External hardware used	N/A	
Performance obtained	N/A	
Driver modules	PL093.c	
Test modules	PL093tst.c	
Standard modules required (normal archive location)	os.c, oss.s, int.c, dispatch.s, boot.s, vectors.s, test.c, main.c	
Custom modules required (in [project]/privates/apPL093	applatfm.h	

3.2 Hardware requirements

3.2.1 FPGA

Is the image used to program the FPGA archived	NO
If so, at what location	N/A

3.2.2 Test configuration

Is the hardware configuration documented	NO
If so, at what location	N/A

3.2.3 External connectors required

none

3.2.4 Devices on prototyping area

none

3.2.5 External test equipment

None

4 CODE COVERAGE

4.1 Overall

Code coverage tests have been performed (generated by v1.18) YES

Code coverage details below

- ◆ Clear the data area below the application base address.
- ◆ Run all test applications.

Percentage of normal routine exits tested. 100 %

- ◆ Examine all routine exits listed as untested to ensure that these are not used in normal flow.

All untested routine exits examined. YES

Percentage of blocks tested. 100 %

4.2 Untested Blocks

Blocks not covered by these tests have been examined and categorised as below.

4.2.1 General cases

The block numbers, as generated by the code coverage tool, are listed below.

Reason	Block numbers
Block is switch statement (end not reached due to break)	
Block end never reached due to break/return statement(s)	
Block implements a trivial range check e.g. if (x<4) {x=4;}	
Default statement in switch not expected to be executed	
Code never executed – included for safety in cases of future expansion.	

4.2.2 Special cases

Block number	Routine	Reason

4.3 Untested Routine Exits

Exit number	Routine	Reason

5 RELEASED TEST MODULE

5.1 Overview

The released test module is named **pl093tst.c**.

It is used as a self-test module for the SSMC PrimeCell and also as a part of the formal testing.

Test environment is as described earlier	YES
Code coverage: Percentage of blocks tested by this test module	100 %

5.2 Changes to test environment

None for public tests.

5.3 Configuration of the test module

The test module is configured through the macro **apPL093_BANK7_BOOT** in system description file **applatfm.h**.

If **apPL093_BANK7_BOOT** is defined in **applatfm.h**, that indicates that system boots from Bank 7, this memory bank is being excluded from the test, and number of banks for test is set up to 7. Otherwise, if **apPL093_BANK7_BOOT** is not defined, Bank 7 is being tested, and number of memory banks is set up to 8.

Macro **apINT_VERSION** in **apintcfg.h** should be defined as follows:

```
# define apINT_VERSION    apVERSION_NONE_ON_LM
```

5.4 Outline

The test module performs the following functions:

5.4.1 Initialisation

- ⊕ Call **apPL093_Initialize()** function to set up peripheral base address.
 - Number of interrupts used 0
 - Specific parameters passed in configuration data NONE
- ⊕ Call **apPL093_StateSizeGet()** function to get size required for storage of the driver state data.

This function is called for test only and return value is not used.

5.4.2 Test Operation

This section is the main body of the test and performs test operations of the driver. After SSMC initialisation it sets up control data of the controller, configures all memory banks with different parameters, reads values of bank registers and then compares them with programmed parameters.

- ⊕ Call **apPL093_ControlSet()** function to set up control data of memory controller.

-
- ⊕ Call apPL093_ControlGet () function to read control data of memory controller.
 - ⊕ Call apPL093_MemBankConfigSet () function to configure memory bank.
 - ⊕ Call apPL093_MemBankConfigGet () function to read configuration parameters of memory bank.
 - ⊕ Call apPL093_InitGet () function to get initialisation flag of the memory bank.
 - ⊕ Call apPL093_WriteEnable () function to enable write operations for the bank.
 - ⊕ Call apPL093_MemBankConfigGet () function to read write protect parameter of the memory bank.
 - ⊕ Call apPL093_WriteDisable () function to disable write operations for the bank.
 - ⊕ Call apPL093_MemBankConfigGet () function to read write protect parameter of the memory bank.
 - ⊕ Call apPL093_ErrorGet () function to read error status flag of the memory bank.
 - ⊕ Call apPL093_StatusGet () function to read state of the memory controller.

5.5 Results

Test run returns TRUE	YES
When apTEST_INTERACTIVE is FALSE, no user interaction is required	YES
When apTEST_VERBOSE is FALSE, no output is produced	YES

5.6 Performance obtained

Performance obtained in this test

```

-----
Testing module PL093 at 0xc0000000
Primecell ID Registers: Pass
Check control data: Pass
Testing Memory Bank #0: Pass  Error Flag: None
Testing Memory Bank #1: Pass  Error Flag: None
Testing Memory Bank #2: Pass  Error Flag: None
Testing Memory Bank #3: Pass  Error Flag: None
Testing Memory Bank #4: Pass  Error Flag: None
Testing Memory Bank #5: Pass  Error Flag: None
Testing Memory Bank #6: Pass  Error Flag: None
Testing Memory Bank #7: Pass  Error Flag: None
Testing External Wait Status: SMWAIT deasserted
apPL093_MemBankConfigSet() error condition invalid parameters : Pass
Testing error condition invalid bank selected : Pass
    *** PASS ***
*** Overall System:  PASS ***

```